

4.3.1 Changes of state and the particle model

4.3.1.1 Density of materials

Content	Key opportunities for skills development
The density of a material is defined by the equation: $\text{density} = \frac{\text{mass}}{\text{volume}}$ $\left[\rho = \frac{m}{V} \right]$ density, ρ, in kilograms per metre cubed, kg/m³ mass, m, in kilograms, kg volume, V, in metres cubed, m³ The particle model can be used to explain • the different states of matter • differences in density.	MS 1a, b, c, 3b, c Students should be able to recall and apply this equation to changes where mass is conserved.

Content	Key opportunities for skills development
Students should be able to recognise/draw simple diagrams to model the difference between solids, liquids and gases.	WS 1.2
Students should be able to explain the differences in density between the different states of matter in terms of the arrangement of atoms or molecules.	WS 1.2

Required practical activity 5: use appropriate apparatus to make and record the measurements needed to determine the densities of regular and irregular solid objects and liquids. Volume should be determined from the dimensions of regularly shaped objects, and by a displacement technique for irregularly shaped objects. Dimensions to be measured using appropriate apparatus such as a ruler, micrometer or Vernier callipers.

AT skills covered by this practical activity: AT 1.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 94).

4.3.1.2 Changes of state

Content	Key opportunities for skills development
Students should be able to describe how, when substances change state (melt, freeze, boil, evaporate, condense or sublimate), mass is conserved.	
Changes of state are physical changes which differ from chemical changes because the material recovers its original properties if the change is reversed.	